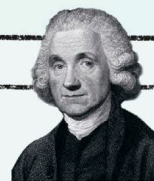


Lavoisier also repeated an experiment carried out by English scientist Henry Cavendish (see pp.94–97), who created a gas he called “inflammable air” when he poured acid onto metals. When Cavendish applied heat to the gas, water was formed. In his experiment, Lavoisier found that the heat acted as a catalyst and made the “inflammable air” (which he renamed “hydrogen”) react more quickly with oxygen in the air to create water. Lavoisier was the first scientist to definitively state that water is made up of hydrogen and oxygen rather than “phlogiston.”

JOSEPH PRIESTLEY

English clergyman Joseph Priestley discovered 10 gases, including “dephlogisticated air,” which Lavoisier named “oxygen.”



Living near a brewery, Priestley (1733–1804) saw something he called “heavy air” (carbon dioxide) bubbling up from the grain and noted it could dissolve in water. He later added mint to a container of carbon dioxide and found it “refreshed” the air sufficiently to support a lighted candle—he had discovered plants emit oxygen. In 1774, Priestley told Lavoisier about his experiments with mercury and “dephlogisticated air” (oxygen). Although his work inspired Lavoisier, Priestley rejected Lavoisier’s theories and continued to believe substances contained “phlogiston.”

**HEATED
MERCURY
OXIDE FOR
12 DAYS
TO DETERMINE
THAT AIR
CONTAINS
OXYGEN**



**HIS TABLE
OF ELEMENTS
INCLUDED OXYGEN
AND HYDROGEN,
BUT ALSO LIGHT
AND HEAT**

